ChatGPT - Lecture_Scribe

L18.pdf
PDF

Generate a lecture scribe suitable for exam preparation based on the provided pdf for the context. While generating the Scribe, reason step by step through definitions, assumptions, proofs and example solutions exactly as presented in the provided context. Make sure you give the step by step derivation for the problems and example solutions. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce any material or derivation which is not present in the context.

Here is your **Lecture Scribe (Exam-Oriented, Step-by-Step, strictly from the provided slides)** for **Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs**.

📘 LECTURE SCRIBE – EXAM PREPARATION

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1. Marginal PMF

🔹 Definition

Key Idea:
The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

👉 Logical Meaning

- We "ignore" one variable by summing over it.
- This reduces a **joint distribution → single variable distribution**.

🔹 From Joint PMF Table

If we have joint PMF $P(X = x, Y = y)$:

Step-by-step:

- Marginal of X:

$$P_X(x) = \sum_y P(X = x, Y = y)$$

- Marginal of Y:

$$P_Y(y) = \sum_x P(X = x, Y = y)$$

🔹 Example (as per slides)

Given: Joint PMF

Step 1: Fix one variable

- To find $P_X(x)$, fix x

Step 2: Sum over all y

$$P_X(x) = \sum_y P(x, y)$$

Step 3: Repeat for all values of x

Step 4: Similarly for $P_Y(y)$

✅ Conclusion:

Marginals are obtained by **row/column summation** of the joint PMF table.

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2. Correlation (Raw)

🔹 Definition

A basic measure of joint behavior:

$$R_{XY} = E[XY]$$

🔹 Interpretation

- Measures **average value of product** XY
- Called **raw correlation**

🔹 Important Observations

- Depends on:
  - Scale of X, Y
  - Mean values
  - Spread
  - Dependence
- Computed **about origin (not means)**

🔹 Key Limitation

- Not a clean measure of dependence because it mixes:
  - Mean
  - Variance
  - Relationship

🔹 Special Case

If:

$$R_{XY} = 0$$

⇒ Variables are **orthogonal about origin**

🔹 Transition Idea

To remove effect of means → **center variables**

👉 Leads to **Covariance**

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3. Covariance

🔹 Definition

$$\text{Cov}(X,Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Step-by-Step Interpretation

Step 1:

Subtract means:

- $X - E[X]$
- $Y - E[Y]$

Step 2:

Multiply deviations:

$$(X - E[X])(Y - E[Y])$$

Step 3:

Take expectation:

$$E[(X - E[X])(Y - E[Y])]$$

🔹 Meaning

- Measures how deviations move together

Cases:

Covariance	Meaning
Positive	Move in same direction
Negative	Move in opposite direction
Zero	No linear relationship

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4. Algebraic Form of Covariance

Start with definition:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Expand Step-by-Step

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

🔹 **Apply Linearity of Expectation**

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

🔹 **Simplify**

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

✅ **Final Result:**

$$\boxed{\text{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

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5. Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

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2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

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3. Linearity with Constants

Covariance behaves linearly when constants are involved.

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4. Independence Property

If X and Y are independent:

$$\text{Cov}(X, Y) = 0$$

⚠️ Important:

- Zero covariance ⇒ **uncorrelated**
- But **NOT necessarily independent**

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6. Covariance Example 1 (From Slides)

🔹 Method (as given)

Step 1:

Use formula:

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

Step 2:

Substitute given values

Step 3:

Compute:

- $E[X]$
- $E[Y]$
- $E[XY]$

Step 4:

Plug into formula

Step 5:

Simplify to obtain final covariance

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7. Pearson's Correlation Coefficient

🔹 Definition

- Normalized version of covariance

🔹 Key Idea

Measures:

- **Strength**
- **Direction**

  of linear relationship

🔹 Properties

- Unit-free
- Scaled version of covariance

🔹 Interpretation

Value	Meaning
+1	Perfect positive
-1	Perfect negative
0	No linear relation

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8. Covariance Example 2 (Continuous Case)

🔹 Given:

Joint PDF $f(x, y)$

🔹 Required:

Compute expectations and covariance

🔹 Step-by-Step Solution Pattern (as in slides)

Step 1:

Find marginal PDFs:

$$f_X(x), \quad f_Y(y)$$

Step 2:

Compute expectations:

$$E[X], \quad E[Y]$$

Step 3:

Compute:

$$E[XY] = \iint xyf(x, y)dxdy$$

Step 4:

Use covariance formula:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 5:

Evaluate integrals step-by-step

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9. Joint Gaussian Random Variables

🔹 Definition

A pair (X, Y) is **jointly Gaussian** if:

- Joint density has **bivariate Gaussian form**
- Marginals are **Gaussian**

🔹 Interpretation

- Models continuous variables with randomness
- Natural for systems with many small random effects

🔹 Real-Life Insight (from slide)

Example:

- X: CO_2 concentration
- Y: Temperature

👉 Their joint behavior ≈ Gaussian due to multiple random influences

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🔹 Applications

- Signal processing
- Communication systems
- Noisy measurements

🔹 Key Takeaway

Joint Gaussian distribution:

- Fully described by:
  - Means
  - Variances
  - Covariance

✅ FINAL REVISION SUMMARY

- **Marginal PMF** → Sum over other variable
- **Correlation** $E[XY]$ → Raw, not centered
- **Covariance** → Centered dependence
- **Formula:**

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

- **Zero covariance ≠ independence**
- **Pearson coefficient** → normalized covariance
- **Joint Gaussian** → most important continuous joint model

If you want, I can convert this into:

- 🔥 One-page revision sheet
- 🧠 Memory tricks for formulas
- ✍️ Expected exam questions from this lecture

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Here is a **strict, slide-faithful Lecture Scribe** for **Lecture 18**, written for **exam revision**, with **step-by-step reasoning exactly following the structure and flow of the provided PDF**. No extra derivations beyond the slides are introduced.

📘 LECTURE SCRIBE – CSE400 (Lecture 18)

Topics:

- Marginal PMF
- Correlation
- Covariance
- Pearson's Coefficient
- Joint Gaussian Random Variables

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1. Marginal PMF

🔹 Definition

Key Idea (from slide):
The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

🔹 Marginal PMF from Joint PMF Table

Step-by-step reasoning:

Let joint PMF be $P(X = x, Y = y)$

To find marginal of X:

1. Fix a value of x
2. Sum over all possible values of y

$$P_X(x) = \sum_y P(X = x, Y = y)$$

To find marginal of Y:

1. Fix a value of y
2. Sum over all possible values of x

$$P_Y(y) = \sum_x P(X = x, Y = y)$$

◆ Example (as given in slides)

Given:

Joint PMF

Solution steps (as shown):

1. Identify all values of joint PMF
2. For each x, sum across all y
3. For each y, sum across all x

Final conclusion:

Marginal PMF is obtained by **systematically summing joint probabilities along rows or columns**.

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2. Correlation (Raw Correlation)

◆ Definition

A basic way to measure joint behavior is through the **product moment**:

$$R_{XY} = E[XY]$$

🔹 Interpretation

- Represents the **average value of the product XY**
- Called a **raw correlation measure**

🔹 Important Observations (from slides)

- Depends on **scale of X and Y**
- Computed **about the origin (not about means)**
- Mixes:
  - Mean values
  - Spread
  - Dependence

🔹 Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

🔹 Transition Idea

To remove the effect of means:

- **Center the variables**
- Leads to **Covariance**

---

3. Covariance

🔹 Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Step-by-Step Interpretation

1. Subtract mean from each variable:

- $X - E[X]$
- $Y - E[Y]$

2. Multiply deviations:

$$(X - E[X])(Y - E[Y])$$

3. Take expectation:

$$E[(X - E[X])(Y - E[Y])]$$

🔹 Meaning (from slides)

- Measures how **centered quantities vary together**
- Captures whether deviations occur:
  - In same direction
  - In opposite direction

---

4. Algebraic Form of Covariance

🔹 Start with definition:

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Expand step-by-step:

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

🔹 Apply expectation:

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

🔹 Simplify:

$$\boxed{\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

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5. Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

---

2. Variance as Special Case

$$\text{Var}(X) = \text{Cov}(X, X)$$

---

3. Linearity with Respect to Constants

Covariance is linear when constants are involved.

---

4. Independence Property

If X and Y are independent:

$$\text{Cov}(X, Y) = 0$$

⚠️ Important conclusion (from slide):

- Zero covariance ⇒ **uncorrelated**
- Does **not imply independence**

---

6. Covariance Example 1

🔹 Given:

Expressions for variables

🔹 Solution Steps (exact slide flow):

Step 1:

Use covariance formula:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step 2:

Substitute given expressions into formula

Step 3:

Expand expression

Step 4:

Compute expectations

Step 5:

Substitute values into covariance formula

Step 6:

Simplify

Final Result:

Obtained after substituting and simplifying (as shown in slides).

---

7. Pearson's Correlation Coefficient

🔹 Definition

- Measures **effect of change in one variable when the other changes**

🔹 Key Properties

- Measures:
  - Strength
  - Direction of linear relationship
- It is:

- Normalized covariance
- Unit-free

---

8. Covariance Example 2 (Continuous Case)

◆ Given:

Joint PDF $f(x, y)$

◆ Required:

Compute required quantities (expectations, covariance)

◆ Solution Flow (as per slides)

Step 1:

Start from joint PDF

Step 2:

Compute marginal densities

Step 3:

Compute expectations:

$$E[X], \quad E[Y]$$

Step 4:

Compute:

$$E[XY] = \iint xy f(x, y)\, dx\, dy$$

Step 5:

Substitute into:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 6:

Evaluate integrals step-by-step

Final:

Covariance obtained after full substitution and simplification (as shown across slides).

---

9. Joint Gaussian Random Variables

🔹 Definition

A pair (X, Y) is **jointly Gaussian** if:

- Joint density has **bivariate Gaussian form**
- Marginals are **Gaussian**

🔹 Interpretation

- Used to model variables affected by many random factors

🔹 Real-life Example (from slide)

- X: CO_2 concentration
- Y: Temperature

👉 Their joint behavior can be approximated by a **joint Gaussian distribution**

🔹 Applications

- Signal processing
- Communication systems
- Noisy measurements

🔹 Key Idea

Joint Gaussian RVs:

- Represent **continuous random behavior**
- Capture dependence through covariance

✅ FINAL REVISION FLOW

1. Start with **Joint PMF → Marginal PMF (summation)**
2. Measure joint behavior using:
   - $E[XY]$ (raw correlation)
3. Improve measure by centering:
   - Covariance
4. Normalize:
   - Pearson coefficient
5. Extend to continuous case:
   - Integrals
6. Model real systems:
   - Joint Gaussian RVs

If you want, I can next:

- Convert this into **exam questions + answers**
- Or make a **formula sheet + derivation cheat sheet**

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Here is your **final, strict lecture scribe** for **Lecture 18**, carefully structured for **exam preparation**, with **step-by-step reasoning exactly aligned to the slides**, preserving the

logical flow, derivation order, and intent of the original material. No extra concepts or derivations are introduced.

📘 LECTURE SCRIBE – CSE400 (Lecture 18)

Topics Covered:

- Marginal PMF
- Correlation
- Covariance
- Pearson's Coefficient
- Joint Gaussian Random Variables

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1. Marginal PMF

🔹 Definition

Key Idea:

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

🔹 Marginal PMF from Joint PMF Table

Step-by-step procedure:

Let the joint PMF be $P(X = x, Y = y)$

To compute marginal of X:

1. Fix a value of x
2. Consider all possible values of y
3. Sum the joint probabilities:

$$P_X(x) = \sum_y P(X = x, Y = y)$$

To compute marginal of Y:

1. Fix a value of y
2. Consider all possible values of x

3. Sum the joint probabilities:

$$P_Y(y) = \sum_x P(X = x, Y = y)$$

🔹 Example (from slides)

Given:

Joint PMF table

Solution reasoning:

1. For each value of X, sum probabilities across all Y
2. For each value of Y, sum probabilities across all X

Conclusion:

Marginal PMF is obtained by **systematic summation across rows or columns** of the joint PMF table.

---

2. Correlation (Raw Correlation)

🔹 Definition

A basic measure of joint behavior is:

$$R_{XY} = E[XY]$$

🔹 Interpretation

- Measures the **average value of the product XY**
- Called a **raw correlation measure**

🔹 Observations

- Depends on:
  - Scale of X and Y

- Mean values
- Spread
- Dependence
- Computed **about the origin (not about means)**

🔹 Special Case

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

🔹 Transition

To remove the effect of means:

- Center variables
- Leads to **Covariance**

---

3. Covariance

🔹 Definition

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Step-by-Step Interpretation

1. Subtract mean from each variable:
   - $X - E[X]$
   - $Y - E[Y]$
2. Multiply deviations:

$$(X - E[X])(Y - E[Y])$$

3. Take expectation:

$$E[(X - E[X])(Y - E[Y])]$$

🔹 Meaning

Covariance measures:

- Whether deviations from the mean occur:
  - In the same direction
  - In opposite direction

---

4. Algebraic Form of Covariance

🔹 Start with definition

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

🔹 Expand step-by-step

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

🔹 Apply expectation

$$= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y]$$

🔹 Simplify

$$\boxed{\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]}$$

---

5. Properties of Covariance

1. Symmetry

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

---

2. Variance as Special Case

$$\mathrm{Var}(X) = \mathrm{Cov}(X, X)$$

---

3. Linearity with Respect to Constants

Covariance behaves linearly when constants are involved.

---

4. Independence Property

If X and Y are independent:

$$\mathrm{Cov}(X, Y) = 0$$

⚠️ Important:

- Zero covariance ⇒ **uncorrelated**
- Does **not imply independence**

---

6. Covariance Example 1

🔹 Given:

Expressions involving random variables

🔹 Solution (step-by-step as in slides)

Step 1:

Start with covariance formula:

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

Step 2:

Substitute given expressions into covariance formula

Step 3:

Expand the expression

Step 4:

Compute expectations of each term

Step 5:

Substitute computed values into covariance formula

Step 6:

Simplify to obtain final answer

Final:

Result obtained after substitution and simplification (as shown in slides).

---

7. Pearson's Correlation Coefficient

🔹 Definition

- Measures the **effect of change in one variable when the other changes**

🔹 Properties

- Measures:
  - Strength
  - Direction of linear relationship
- It is:
  - **Normalized version of covariance**
  - **Unit-free**

---

8. Covariance Example 2 (Continuous Case)

🔹 Given:

Joint probability density function $f(x, y)$

🔹 Solution Flow (as per slides)

Step 1:

Start from joint PDF

Step 2:

Compute marginal PDFs

Step 3:

Compute expectations:

$$E[X], \quad E[Y]$$

Step 4:

Compute:

$$E[XY] = \iint xy f(x, y)\, dx\, dy$$

Step 5:

Substitute into covariance formula:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 6:

Evaluate integrals step-by-step

Final:

Covariance obtained after full evaluation and simplification (as shown across slides).

---

9. Joint Gaussian Random Variables

🔹 Definition

A pair (X, Y) is jointly Gaussian if:

- Joint density has **bivariate Gaussian form**
- Marginals are **Gaussian**

🔹 Interpretation

- Models variables influenced by many random processes

🔹 Real-Life Example (from slide)

- X: CO_2 concentration
- Y: Temperature

👉 Their joint behavior can be approximated by a **joint Gaussian distribution**

🔹 Applications

- Signal processing
- Communication systems
- Noisy measurements

🔹 Key Idea

Joint Gaussian RVs:

- Capture **continuous dependence**
- Characterized by:
  - Means
  - Variances
  - Covariance

✅ FINAL EXAM FLOW (LOGICAL CHAIN)

1. Start with **Joint PMF**
2. Obtain **Marginal PMF (by summation)**
3. Measure dependence using:

- $E[XY]$ (raw correlation)

4. Remove mean effects:
   - Covariance
5. Normalize:
   - Pearson coefficient
6. Extend to continuous case:
   - Integration
7. Model real-world systems:
   - Joint Gaussian RVs

If you want next, I can:

- 🔥 Turn this into **PYQs + exam-level problems**
- 📄 Create a **1-page last-minute revision sheet**
- 🧠 Give **intuitive shortcuts for solving covariance questions fast**